VEILLE — Flaws, Risks & Proposed Solutions

Comprehensive technical review of the updated repository/codebase and empirical-validation walkthrough

Repository state: main branch, commit 5b8a305

Objective: identify remaining technical, scientific, provenance, reproducibility, security and documentation weaknesses, then define concrete fixes before zero-shot external evaluation and targeted training.

Area	Assessment	Priority
Architecture	Strong; retain overall design	—
Canonical adapters	Strong and scalable	P1
Dataset provenance	Improved; ICIJ/Enron still need correction	P0
Entity Resolution	Evaluation improved but does not fully exercise production ER	P0
GraphRAG	Evaluation improved but claim support remains a proxy	P0
NER	Recovery metric useful; add exact-span evaluation	P1
Reproducibility	Needs experiment manifests and async completion checks	P1
Documentation	Improved; root README still needs alignment	P1

1. P0 — Production ER evaluator mismatch

Flaw / risk: The new evaluator correctly uses labeled candidate pairs and a confusion matrix, but it calculates its own lexical similarity rather than invoking the production VEILLE ER scoring path. The 100% auto-merge precision therefore should not yet be presented as production ER accuracy.

Solution: Pass every labeled pair through the same production ER function used by the review queue. Record lexical, structural and combined scores, thresholds, decision and ground-truth label. Compute TP/FP/TN/FN/HITL from those production decisions.

2. P0 — ER benchmark is too small

Flaw / risk: Twenty-four labeled pairs are useful for controlled validation but insufficient for a strong generalization claim.

Solution: Build a larger labeled benchmark (roughly 500–2,000 pairs) covering exact matches, formatting noise, OCR errors, aliases, transliteration, abbreviations, common-name collisions, type mismatches, address variation and adversarial lookalikes. Report confidence intervals where useful.

3. P0 — Enron metadata is not yet canonical

Flaw / risk: The updated code removes the old hard-coded participant assignment and attempts header extraction, but it uses a derived Enron spam/ham source and still has generated fallback identities. This can create synthetic communication metadata.

Solution: Use the original Enron mailbox/email corpus. Parse actual From/To/Cc/Bcc/Date/Subject/Message-ID headers. If participant metadata is absent, preserve it as missing and do not invent identities.

4. P0 — ICIJ acquisition provenance is incomplete

Flaw / risk: The manifest hashes a local ICIJ slice, but the downloader does not establish the same explicit official-source download chain as the other datasets.

Solution: Download from the declared official source, preserve the original artifact, compute original SHA-256, apply deterministic selection, compute derived-corpus SHA-256 and record source/release/date/selection metadata.

5. P0 — GraphRAG claim support remains a proxy

Flaw / risk: The new evaluator is much better than keyword markers, but a sentence can be treated as supported when it contains a known graph node. Entity presence does not prove that the entire factual assertion is supported.

Solution: Decompose claims into subject–predicate–object plus amount/time/location where relevant. Verify the relation in Neo4j and then verify supporting evidence in PostgreSQL. Classify claims as SUPPORTED, PARTIALLY_SUPPORTED, UNSUPPORTED or CONTRADICTED.

6. P0 — Citation validity ≠ citation entailment

Flaw / risk: A 100% citation-verification score can show that evidence IDs exist, but not that the cited evidence supports the claim.

Solution: Report Citation Validity and Citation Entailment separately. Every factual claim should link to evidence and be checked for support.

7. P1 — NER is not standard span-level evaluation

Flaw / risk: Current end-to-end recovery relies on name presence/containment. That is useful, but it is not standard exact-span NER evaluation.

Solution: Add character/token span evaluation with exact-span P/R/F1 and per-label scores. Keep the existing recovery metric as a separate end-to-end diagnostic.

8. P1 — Taxonomy mapping needs versioning

Flaw / risk: InLegalNER has a richer label taxonomy than the canonical VEILLE model; the transformation should be explicit.

Solution: Create a versioned taxonomy_mapping YAML/JSON: source label, canonical label, transformation rule and whether information is lossy. Version it with the adapter.

9. P1 — Fixed sleep harms reproducibility

Flaw / risk: Benchmark synchronization relies on a fixed delay, which can evaluate partially processed data on slower environments.

Solution: Poll ingestion/job status. Require successful completion, settled outbox state and stable graph counts before evaluation; fail on timeout.

10. P1 — Benchmark isolation/cleanup

Flaw / risk: Repeated runs can contaminate persistent graph/database state.

Solution: Use a unique benchmark run ID, isolate all records, export results, then clean up the benchmark namespace or use a dedicated benchmark database/schema.

11. P1 — Missing experiment provenance

Flaw / risk: Dataset manifests identify source artifacts, but experiment results also depend on Git commit, model version, thresholds and configuration.

Solution: Create immutable experiment manifests containing experiment ID, Git commit, dataset hashes, adapter/model versions, configuration, environment and metric artifacts.

12. P1 — Observed vs inferred relationships

Flaw / risk: A directly observed evidence edge is materially different from an algorithmically inferred or analyst-confirmed relationship.

Solution: Add explicit states such as OBSERVED, INFERRED, ANALYST_CONFIRMED and REJECTED. Expose them in UI and GraphRAG context.

13. P1 — Relationship merge fallback can lose semantics

Flaw / risk: The native Cypher fallback can collapse relationship types into ASSOCIATED_WITH while preserving the old type only as a property.

Solution: Recreate original relationship types during merge or make semantic-preserving APOC behavior a required dependency. Add regression tests for all important relation types.

14. P1 — GraphRAG retrieval is keyword-centric

Flaw / risk: Keyword matching plus fixed node/edge limits can miss semantically relevant context at scale.

Solution: Use staged retrieval: entity retrieval → entity linking → graph expansion → relationship ranking → evidence ranking → context-budget selection, with temporal and case constraints.

15. P1 — Source/import/entity counts need strict terminology

Flaw / risk: Source volume, selected records, parsed records and canonical graph nodes/edges are different quantities.

Solution: Standardize manifest fields: source_record_count, selected_record_count, parsed_record_count, rejected_record_count, canonical_entity_count and canonical_relationship_count.

16. P1 — Real/synthetic/controlled taxonomy must propagate

Flaw / risk: IBM AMLWorld and Operation Storm Watch are valuable but must never be confused with real-world evidence.

Solution: Add mandatory provenance_class to every evidence/graph object and expose it in UI and GraphRAG prompts.

17. P1 — Documentation has multiple sources of truth

Flaw / risk: Root README and newer codebase documentation can diverge.

Solution: Make the root README authoritative; align versions, infrastructure, terminology, claims and benchmark links.

18. P2 — Avoid unsupported security/legal claims

Flaw / risk: Do not reintroduce claims such as court admissibility, zero data loss, zero hallucination, or production-grade law-enforcement deployment without independent evidence.

Solution: Use research-prototype, tamper-evident SHA-256 integrity, transactional outbox, RBAC and institutional-evaluation language.

19. P2 — Metrics should be machine-generated artifacts

Flaw / risk: Headline benchmark values are safer when generated directly from the experiment rather than manually maintained.

Solution: Produce metrics.json, er_results.json, rag_results.json and benchmark_report.md from one run, all tied to an experiment ID and Git commit.

20. P2 — Critical guarantees need regression tests

Flaw / risk: Provenance propagation, relationship semantics, ER thresholds, case isolation and evidence linkage can regress during refactoring.

Solution: Add automated tests for ER decisions, false-merge safeguards, provenance propagation, relationship preservation, outbox idempotency, benchmark isolation and claim/evidence verification.

21. Target Architecture After Corrections

Stage	Required behavior
Source acquisition	Official source → artifact → original SHA-256
Selection	Deterministic sampling/filtering with recorded methodology
Canonicalization	Source labels mapped through versioned taxonomy
NLP	Exact-span NER + relation extraction diagnostics
Entity Resolution	Production ER engine evaluated against labeled pairs
Graph	Observed/inferred/confirmed states + evidence provenance
GraphRAG	Entity retrieval → graph expansion → evidence ranking
Claim verification	Claim → triple → graph relation → source evidence
HITL	Ambiguous/high-impact decisions routed to human review
Experiment	Git commit + dataset hashes + config + model versions + metrics

22. Priority Roadmap

Phase	Work	Exit criterion
P0 Correctness	Production ER evaluation; structured GraphRAG verification; authentic headers/provenance acquisition	No major Evaluator/actual CI mismatch
P1 Reproducibility	Async completion checks; run isolation; experiment manifests; taxonomy mapping; exact-span NER	Fresh environment can reproduce the experiment definition
P2 Robustness	Large/adversarial ER set; semantic merge tests; stronger retrieval	Failure modes quantified
P3 Zero-shot	Evaluate real external datasets without fine-tuning	Baseline weaknesses identified
P4 Targeted training	Train only weak components and compare on held-out data	Measured before/after improvement

23. Metrics to Report

Subsystem	Metrics
NER	Exact-span Precision / Recall / F1; per-label F1
ER	Precision / Recall / F1; false-merge rate; false-split rate; auto-merge coverage; HITL rate
Relations	Precision / Recall / F1 by relation type
Retrieval	Precision@K; Recall@K; MRR; entity-linking accuracy
GraphRAG	Supported / unsupported / contradicted claims; citation validity; citation entailment
Provenance	Artifact hash verification; source-to-canonical traceability; provenance completeness
System	Ingestion completion; outbox retries/idempotency; benchmark reproducibility

24. Final Strategic Recommendation

Freeze feature development temporarily. VEILLE is already feature-rich enough for the next milestone. Prioritize correctness, provenance, reproducibility and failure-mode measurement.

Do not train yet. After the P0/P1 corrections, run a clean zero-shot baseline. Use the measured weaknesses to decide whether NER, ER, relation extraction or retrieval actually requires targeted training.

Positioning: present VEILLE as a provenance-aware investigative intelligence research prototype that fuses heterogeneous evidence into a canonical graph and produces evidence-grounded analytical leads with human review—not as an autonomous guilt-determination or crime-prediction system.

Prepared from the updated repository review and the uploaded “Academic Review Resolution & Empirical Validation Walkthrough.”